← Back **Learning Check - Logistic Regression**
Graded Assignment • 20 min

English Due Oct 4, 11:59 PM IST

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Introduction to Week 4

Lesson 1: Introduction to Classification

Lesson 2: Logistic Regression

- **Video:** Logistic Regression 4 min
- **Video:** Maximum Likelihood 8 min
- **Graded Assignment:** Learning Check - Logistic Regression Submitted

Lesson 3: Logistic Regression, Continued

Weekly Assignment

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

Learning Check - Logistic Regression

Review Learning Objectives

Coach

Ready to review what you've learned before starting the assignment? I'm here to help.

Help me practice   Let's chat

...mpts every 8 hours)   **Try again**

To pass you need at least 80%. We keep your highest score.

100%

View submission   **See feedback**

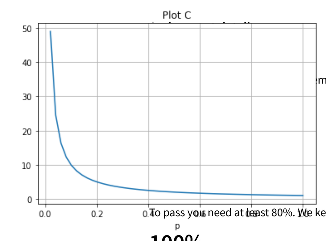
...rt an issue

1. In the following exercise, we will derive the logistic regression function through a series of logical steps. **1 / 1 point**

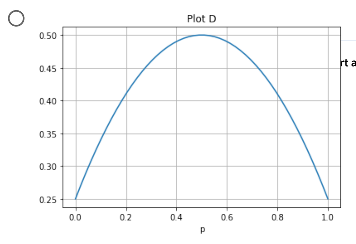
Define the probability function $p(x) = Pr(Y=1|X=x)$. We define the odds as the probability that $Y=1$ given $X=x$ divided by the complementary probability, i.e., the odds are equal to $\frac{p(x)}{1-p(x)}$.

Question: Which of the following graphs represents this function $f(p) = \frac{p}{1-p}$, where the x-axis represents the variable $p \in [0,1]$?

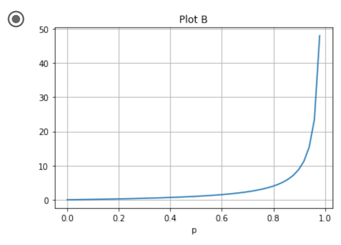
○ Plot C



○ Plot D



◉ Plot B



Correct
Correct

2. Let us define the log-odds (also called logit) function as $ln(\frac{p(x)}{1-p(x)})$. Let us use the logit function as the output variable of a linear regression $Y = ln(\frac{p(X)}{1-p(X)}) = \beta_0 + \beta_1 X$. Based on this last expression, we would like to mathematically derive an explicit formula for $p\,(X)$. **1 / 1 point**

Which of the following choices shows the correct steps to derive a formula for $p\,(X)$? The final answer should express the probability function as a sigmoid.

○
1. $\frac{p(X)}{1-p(X)} = e^{\beta_0+\beta_1 X}$
2. $p(X) \times (1 + e^{\beta_0+\beta_1 X}) = e^{\beta_0+\beta_1 X}$
3. $p(X) = e^{\beta_0+\beta_1 X} - p(X)e^{\beta_0+\beta_1 X}$
4. $p(X) = (1 - p(X)) \times e^{\beta_0+\beta_1 X}$
5. $p(X) = \frac{e^{\beta_0+\beta_1 X}}{1+e^{\beta_0+\beta_1 X}}$

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5. $p(X) = (1 - p(X)) \times e^{\beta_0+\beta_1 X}$

Correct
Correct